

BONNARD, QC, Canada

WMO: 713830

Lat: 50.729N

Lon: 71.013W

Elev: 1634

StdP: 13.85

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-28.0	-39.8	0.6	-32.5	-34.4	0.8	-27.9	26.4	9.1	23.6	1.5	3.6	260	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	78.0	63.0	74.8	61.7	71.8	60.8	66.7	72.8	64.8	71.0	63.0	68.8	11.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	97.8	68.9	62.7	90.5	67.1	60.7	84.2	65.5	32.3	72.9	30.8	71.1	29.4	68.8	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	20.5	18.2	-42.0	84.1	4.1	3.3	-45.0	86.4	-47.4	88.4	-49.7	90.2	-52.7	92.6		
(5)			-41.4	69.9	4.3	3.1	-44.5	72.2	-47.0	74.0	-49.4	75.7	-52.5	78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.1	-2.3	-0.9	11.4	27.2	42.2	55.2	60.5	58.7	50.5	38.0	23.9	6.9
	DBStd	24.37	14.16	13.80	13.84	10.09	7.85	7.59	5.18	5.51	7.31	7.32	9.56	12.86
	HDD50	7808	1622	1425	1196	683	269	31	0	4	80	378	783	1337
	HDD65	12420	2087	1845	1661	1133	709	307	159	208	439	837	1233	1802
	CDD50	909	0	0	0	0	25	186	325	272	94	7	0	0
	CDD65	48	0	0	0	0	1	13	19	12	3	0	0	0
	CDH74	352	0	0	0	0	24	112	133	67	16	0	0	0
	CDH80	40	0	0	0	0	3	16	16	4	1	0	0	0
	WSAvg	7.9	7.6	7.9	8.0	8.1	7.4	7.8	7.5	7.5	8.4	8.8	8.5	7.4
	PrecAvg	39.6	2.9	1.5	1.6	2.4	2.9	4.3	5.4	4.3	4.3	3.0	3.6	3.3
(15) Precipitation	PrecMax	48.3	4.6	3.0	2.9	4.7	6.3	7.9	9.2	6.5	6.0	4.8	5.3	6.2
	PrecMin	30.6	1.1	0.6	0.3	0.7	0.6	2.5	2.4	2.4	2.3	1.5	1.1	1.4
	PrecStd	4.6	1.0	0.7	0.7	1.1	1.2	1.4	1.5	1.2	0.9	1.1	1.1	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.3	38.6	46.3	55.1	77.5	82.2	82.4	79.6	76.0	63.5	47.8	35.8
		MCWB	37.6	36.2	38.5	42.4	59.8	64.8	66.2	65.2	64.8	56.2	45.7	34.7
	2%	DB	32.3	32.9	39.1	49.0	68.3	76.9	77.7	75.6	70.1	56.8	42.8	33.0
		MCWB	31.9	31.1	34.3	39.4	54.1	62.1	62.7	63.5	62.3	52.2	41.0	32.4
	5%	DB	25.4	27.8	35.3	44.3	61.8	73.4	74.4	72.0	66.2	53.2	38.6	29.7
		MCWB	24.5	26.2	32.4	37.5	50.6	60.1	61.6	61.8	59.8	49.8	37.0	29.2
	10%	DB	19.8	21.9	32.2	40.9	56.2	69.4	71.3	69.0	62.5	49.1	36.0	25.6
		MCWB	19.0	20.5	30.1	35.4	47.9	57.9	60.9	60.7	56.8	46.3	34.8	24.9
	0.4%	WB	36.2	37.4	40.2	44.1	62.0	69.1	69.7	68.3	67.3	57.1	47.0	35.1
		MCDB	36.8	37.9	44.0	50.7	73.4	76.5	76.0	74.1	72.3	61.2	47.8	35.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.5	30.8	35.6	40.9	56.2	65.6	66.3	66.1	63.7	53.7	41.6	32.2
		MCDB	32.0	32.7	37.9	47.0	65.4	73.0	72.1	71.4	68.7	56.2	42.2	32.8
	5%	WB	24.0	25.4	32.7	38.5	52.3	62.8	64.3	64.2	60.8	50.1	37.5	29.0
		MCDB	24.7	27.2	34.8	43.1	60.3	69.4	70.9	69.6	64.6	52.3	38.2	29.5
	10%	WB	18.6	20.2	29.8	36.3	48.7	59.8	62.6	62.2	57.8	46.6	34.8	24.8
		MCDB	19.4	21.6	32.2	40.2	55.3	66.6	68.9	67.1	61.6	48.8	35.8	25.4
(35) Mean Daily Temperature Range	5% DB	MDBR	20.1	24.4	25.3	20.9	25.0	19.8	17.8	16.6	15.5	11.0	10.9	16.3
		MCDBR	20.2	21.9	25.1	24.9	31.1	25.8	22.8	21.6	20.5	17.8	12.3	14.8
		MCWBR	19.3	20.7	20.7	16.9	18.6	13.0	11.4	11.6	12.8	13.2	11.2	14.9
	5% WB	MCDBR	19.3	21.4	20.8	20.8	28.1	20.5	18.4	17.4	18.0	15.3	12.0	14.9
		MCWBR	19.2	19.8	17.7	15.0	18.4	12.2	10.6	10.9	13.0	12.4	11.6	15.0
(40) Clear-Sky Solar Irradiance	taub		0.245	0.254	0.281	0.326	0.340	0.368	0.378	0.360	0.340	0.309	0.283	0.241
	taud		2.266	2.267	2.238	2.231	2.432	2.403	2.390	2.453	2.487	2.543	2.416	2.302
	Ebn at Noon		253	283	293	290	291	282	277	277	271	262	238	238
	Edh at Noon		21	28	35	40	34	36	36	32	28	22	19	17
(44) All-Sky Solar Radiation	RadAvg		379	734	1166	1524	1710	1720	1556	1259	948	540	275	223
	RadStd		43	63	102	115	123	139	137	78	72	54	30	32

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page